Digital Transformation in Higher Education: Evaluating the Digital Competence Performance of Academics Using WENSLO and RAWEC Methods

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Abstract. Digital transformation is a process that requires individuals and institutions to adapt to technological change, improve their competitive advantage and plan strategically for sustainable development. In this context, higher education institutions play a crucial role in promoting digital transformation. Therefore, this study was conducted to assess the digital competencies of academics in order to increase efficiency in higher education and support the digitalization process. The six main areas defined by DigCompEdu were used to analyze the digital competences of academics. The study included academic staff from faculties in three technical and three health-fields. The data on digital competencies was collected through self-assessment via a survey. The data acquired from this survey was analyzed using the WENSLO method to calculate the weighting of each criterion. In order to evaluate the performance of the faculties digital competence, the best performing faculties were ranked using the RAWEC method. The results of the analyses show that the criterion Learners' Digital Competence with the highest weighting value has a greater influence on the faculty-based evaluation of digital competences, while the criterion Teaching and Learning with lowest criterion weighting has a lesser influence. Furthermore, the results show that faculties in the technical field have higher digital competences than faculties in the health field. This study helps to identify priority areas for the development of digital competences at universities and provides valuable insights for the strategic planning of digital transformation processes.

Keywords: Digital Competence · Higher Education · Digital Transformation · WENSLO Method · RAWEC Method

1 Introduction

Digital transformation refers to the transformation processes in society and industry through digital technologies, infrastructures and platforms. The use of digital-based information, communication and technology goals to improve society and businesses.

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This concept generally requires organizations to develop innovative strategies to create more efficient and effective operational processes using digital technologies [1]. This transformation process is of great importance not only for businesses, but also for higher education institutions. Higher education institutions can make their educational processes more efficient by using digital technologies and innovative strategies. But as digital technologies began to play an unavoidable role in our daily lives, this led to both changes and challenges. This has led to the notion of digital competence taking a central place in our lives. As a result, digital skills have rapidly become a core competence area and an important topic in various policy documents and strategies [2].

Educators working in higher education institutions should have digital competences in order to educate individuals according to the requirements of the age. Digital competence refers to being able to employ technology and digital tools successfully in a professional context. This includes the ability to teach with digital tools, manage digital learning environments and work with other digital skills [3]. There are various frameworks for measuring digital competences. The European Union Commission created the Digital Competence Framework for Educators to test and enhance teachers' and educators' digital competence. As a result of this work, the 'DigCompEdu Framework' was created [4, 5]. DigCompEdu is one of the most complete and up-to-date models for today's educators, with 22 competences structured into six primary dimensions. These dimensions are professional engagement, digital resources, teaching and learning, assessment, empowering learners facilitating, learners' digital competence [5].

The aim of this study is to investigate the perceptions of academics in six faculties of a major state university in Turkey regarding their digital competencies and to evaluate how these competencies develop in different dimensions at the faculty level. A survey has been prepared using the DigCompEdu methodology for evaluating academics' digital competencies.

The weights of the academics in the digital competence dimensions were calculated using the Weights by ENvelope and SLOpe (WENSLO) method. In addition, the Ranking Alternatives with Weights of Criterion (RAWEC) method was used to analyze the digital competences between faculties and the digital competence levels of each faculty were assessed. This approach reveals which faculties have stronger digital competences that affect academic success and which faculties need to improve in this area. As a result of this assessment, the faculties with the highest digital competences was identified and the study aims to contribute to the development of policies and strategies to increase digital competences. The originality of this study lies in the assessment of digital competences based on the DigCompEdu framework and the analysis of the results with multi-criteria decision making (MCDM) methods, which are not widely used in this field. As far as the authors knowledge, no study has been found that has used the DigCompEdu framework and MCDM methods together to identify faculties digital competencies.

The paper is organized as follows: Sect. 2 presents a review of relevant literature. Section 3 outlines the research methodology. Section 4 provides a detailed case study. Lastly, Sect. 5 concludes with a summary of key findings.

2 Literature Review

The DigCompEdu framework has been used in various studies to assess and improve the digital competences of educators. In Latin America, teachers' digital competences were found to be at a medium level and training plans were needed [6]. A research in Portugal found that gender, age, and experience had no significant effect on digital competence, but the framework was beneficial for identifying weaknesses [7]. In the Spain-Peru comparison, the importance of personalized training programmes for the development of digital competences was emphasized [8], while in the Spain-Portugal comparison, the digital competences of Portuguese teachers were found to be higher [9]. The Germany study discovered that digital competence is based on the idea that it advances via experience and diversification of digital strategies [10]. These studies reveal the applicability of DigCompEdu in different educational contexts and digital competence development strategies.

3 Methodology

3.1 WENSLO Method

The WENSLO method, an objective method for weighting criteria, was developed by Pamucar et. al. [11]. The method takes into account the ratio between the envelope value and the slope value of each criterion. When the envelope value is high and the slope value is low, a higher weight is given to that criterion [12]. The method is low computational complexity and easy to understand. WENSLO method steps are given below [11].

Step 1. Creating the decision matrix

$$\left[\zeta_{ij}\right]_{mxn} = \begin{bmatrix} A/C & C_1 & C_2 & \cdots & C_j \\ Target & maxmin & maxmin & \cdots & maxmin \\ A_1 & \zeta_{11} & \zeta_{12} & \cdots & \zeta_{1j} \\ A_2 & \zeta_{21} & \zeta_{22} & \cdots & \zeta_{2j} \\ \cdots & \cdots & \cdots & \ddots & \cdots \\ A_i & \zeta_{i1} & \zeta_{i2} & \cdots & \zeta_{mn} \end{bmatrix} \tag{1}$$

where, m, alternative; n, number of criteria.

$A_1, A_2, ..., A_m$ the vector space of alternatives; $C_1, C_2, ..., C_n$ the vector space of criteria.

maxmin, depending on the objective towards which the criterion is orientated, in a benefit environment the objective of the criterion is to achieve maximum value, while in a cost environment the objective of the criterion is to achieve minimum value.

ζ_{ij} is the estimated value of the i'th alternative according to the j'th criterion.

Step 2. Normalization.

$$z_{ij} = \frac{\zeta_{ij}}{\sum_{i=1}^{m} \zeta_{ij}} \forall_j \in [1,\ 2,\ \ldots,\ n] \tag{2}$$

z_{ij}, represents the elements of the normalized decision matrix and must have values between 0 and 1.

Step 3. Calculation of criterion class range

$$\triangle z_j = \frac{\max\limits_{1<i\leq m} z_{ij} - \min\limits_{1<i\leq m} z_{ij}}{1 + 3.322 \times \log m}$$

$$\triangle z_j = \frac{\max\limits_{1<i\leq m} z_{ij} - \min\limits_{1<i\leq m} z_{ij}}{1 + 3.322 \times \log m} \tag{3}$$

Step 4. Calculation of the criterion slope

$$tan\varphi_j = \frac{\sum_{i=1}^{m} z_{ij}}{(m-1) \cdot \Delta z_j} \forall_j \in [1, 2, ..., n] \tag{4}$$

Step 5. Determination of the criterion envelope

$$E_j = \sum\nolimits_{i=1}^{m-1} \sqrt{\left(z_{i+1,j} - z_{i,j}\right)^2 + \triangle z_j^2}\ \sqrt{\forall_j \in [1, 2, ..., n]} \tag{5}$$

Step 6. Defining the envelope-slope ratio

$$q_j = \frac{E_j}{tan\varphi_j} \forall_j \in [1, 2, ..., n] \tag{6}$$

Step 7. Calculation of criterion weights

$$w_j = \frac{q_j}{\sum_{j=1}^{n} q_j} \forall_j \in [1, 2, ..., n] \tag{7}$$

3.2 RAWEC Method

The RAWEC method, which was introduced by Puska et. al., is used for ranking alternatives in order to simplify decision-making processes. While other alternative ranking methods complicate the process with additional steps, RAWEC consists of only four steps and the first two steps are common to all methods. The uniqueness of this method lies in the fact that it combines the construction of a weighted normalized decision matrix with the calculation of the deviation of ideal and anti-ideal values. The steps of the method are given below [13].

Step 1. Creating the decision matrix

$$\mathcal{X} = \begin{matrix} \\ A_1 \\ A_2 \\ \vdots \\ A_m \end{matrix} \begin{bmatrix} C_1 & C_2 & . & C_n \\ x_{11} & x_{21} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m1} & \cdots & x_{mn} \end{bmatrix} \tag{8}$$

Step 2. Normalization of the decision matrix.
Double normalization is done by applying n_{ij} and $n_{ij}\prime$ successively.

$$n_{ij} = \frac{x_{ij}}{x_{j\,max}}, \textit{ and } n_{ij}\prime = \frac{x_{j\,min}}{x_{ij}}, \textit{ for benefit criteria} \tag{9}$$

$$n_{ij} = \frac{x_{j\,min}}{x_{ij}}, \textit{ and } n_{ij}\prime = \frac{x_{ij}}{x_{j\,max}}, \textit{ for cost criteria} \tag{10}$$

where x_{jmin} is the lowest value of the alternatives according to the criteria and, x_{jmax} is the highest value of the alternatives.

Step 3. Calculation of deviation from criterion weight

$$v_{ij} = \sum_{i=1}^{n} w_j \cdot \left(1 - n_{ij}\right) \tag{11}$$

$$v_{ij}\prime = \sum_{i=1}^{n} w_j \cdot \left(1 - n'_{ij}\right) \tag{12}$$

Step 4. Calculation of the value of the RAWEC method

$$Q_i = \frac{v'_{ij} - v_{ij}}{v'_{ij} + v_{ij}} \tag{13}$$

Q_i, ranges from -1 to 1. The ideal option is the one with the highest value.

4 Case Study

In the study, the degree of importance of each dimension was determined using the WENSLO method, one of the MCDM methods. MCDM methods provide clearer, consistent and high-quality results by evaluating multiple factors simultaneously [14]. RAWEC method was utilized to rank the digital competence performance of faculties. This method was used to compare the digital competence levels of the faculties and to determine which faculty is stronger in terms of digital competence. These methods make it easier to make optimal decisions in complex evaluation processes and to compare different options more effectively. Figure 1 shows the study's hierarchical structure.

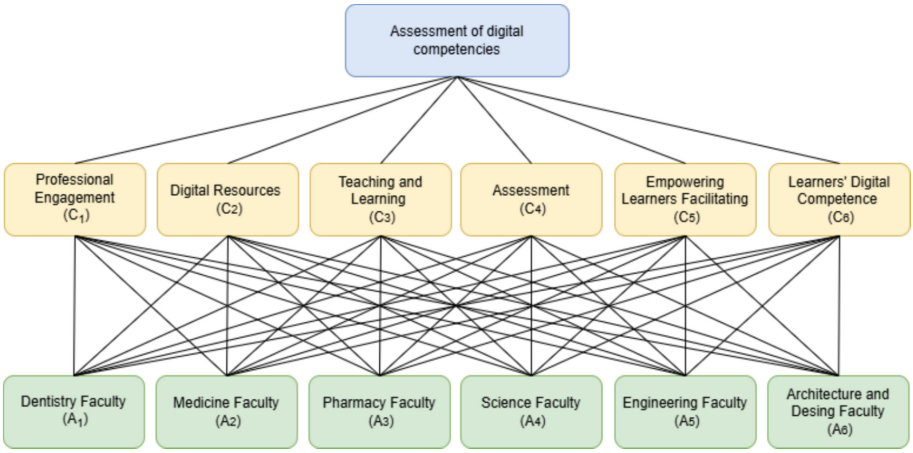


Fig. 1. Hierarchical structure

4.1 Sample and Data Collection

A total of 246 academic staff from three technical and three health faculties at university participated in the study via a survey. The assessment tool 'DigCompEdu Check-In Tool', which was developed as part of the European Digital Competence Framework for Educators, was used to analyze the digital competences of lecturers teaching in the faculties of Engineering, Science, Architecture, Pharmacy, Dentistry and Medicine [5, 15]. In the study, the items of the Turkish version of the assessment tool were adapted to higher education and used. [4, 16]. There are 22 multiple-choice questions of the 5-point Likert type (1: Strongly Disagree…5: Strongly Agree) to determine the digital competences of academics. Each question in the data collection tool is given a score between 0 and 4. In this survey, a 6-dimensional scale was used to measure digital competences. In order to balance the effect of the different number of questions between the dimensions and to balance the effect of the different number of participants in the faculties, the weighted average of the survey results was taken. The reliability of the internal consistency of the assessment instrument was calculated using Cronbach's alpha reliability analysis and resulted in a value of 0.941.

4.2 Implementation of the WENSLO Method

The initial decision matrix in Eq. (1) was formed using the values obtained from the survey results. The initial decision matrix was then normalized using Eq. (2) and presented in Table 1.

Table 1. Normalized decision matrix

A/C	C_1	C_2	C_3	C_4	C_5	C_6
A_1	0.1574	0.1642	0.1618	0.1518	0.1562	0.1608
A_2	0.1659	0.1647	0.1591	0.1505	0.1454	0.1478
A_3	0.1709	0.1709	0.1712	0.1663	0.1715	0.1684
A_4	0.1682	0.1689	0.1634	0.1725	0.1642	0.1604
A_5	0.1637	0.1739	0.1726	0.1741	0.1764	0.1747
A_6	0.1738	0.1574	0.1719	0.1848	0.1862	0.1877

Using Eqs. (3)-(6) Δz_j, $tan\varphi_j$, E_j, q_j are calculated and given in Table 2.

Table 2. Calculation results

	C_1	C_2	C_3	C_4	C_5	C_6
$\triangle z_j$	0.0046	0.0046	0.0038	0.0095	0.0114	0.0111
$tan\varphi_j$	43.7206	43.6074	53.0963	20.9489	17.6081	17.9634
E_j	0.0393	0.0412	0.0397	0.0634	0.0892	0.0895
q_j	0.0009	0.0009	0.0007	0.0030	0.0051	0.0050

Final criteria weights were calculated using Eq. (7) and presented in Table 3.

Table 3. Final criteria weights

	C_1	C_2	C_3	C_4	C_5	C_6
w_{ij}	0.0574	0.0603	0.0477	0.1933	0.3233	0.3181

According to the results, C_6 has the highest weight, C_5 has the second highest weight and C_3 has the lowest weight and C_1 have the second lowest weight.

4.3 Implementation of RAWEC Method

By following the steps of the RAWEC method, the ranking of the alternatives was determined and presented in Table 4.

Table 4. Ranking of alternatives

Alternatives	v_{ij}	v'_{ij}	Q_i	Rank
A_1	0.1431	0.0523	-0.4646	5
A_2	0.1821	0.0059	-0.9368	6
A_3	0.0790	0.1185	0.2002	4
A_4	0.1023	0.7225	0.7520	2
A_5	0.0530	0.1427	0.4584	3
A_6	0.0062	0.1815	0.9336	1

According to the results, the Faculties of Architecture (A_6), Science (A_4) and Engineering (A_5) are the faculties with the highest digital competence, while the Faculties of Medicine (A_2), Dentistry (A_1) and Pharmacy (A_3) are ranked as faculties with less digital competence.

5 Conclusion

The assessment of digital competences is important in terms of creating important opportunities and development areas for both individuals and organizations. Success in the digital world can be achieved by correctly identifying digital competences and developing the right strategies according to these competences. In this study, the digital competences of academics working in six selected faculties of a university in Turkey were evaluated. The results obtained from the academics by means of a questionnaire were analyzed with the MCDM methods WENSLO and RAWEC.

The results of the WENSLO method show that Learners' Digital.

Competence (C_6) and Empowering Learners Facilitating (C_5) are the most important criteria in the decision process, while Teaching and Learning (C_3) and Professional Engagement (C_1) are the least important criteria. This means that the results of criteria C_6 and C_5 are the most influential and play a decisive role in the final decision. According to the results of the RAWEC method, it is concluded that the digital competences of academics in the faculties of Architecture (A_6), Science (A_4) and Engineering (A_5) are better than those in the faculties of Dentistry (A_1), Medicine (A_2) and Pharmacy (A_3). In other words, the technical faculties perform better than the health faculties. This situation shows that academics at technical faculties accept the development of digital competences as a professional requirement. These academics use the ability to work effectively with digital tools, not only in teaching, but also in research and project development. Academics in health faculties tend to see digital competences as an additional qualification. This could be due to the fact that academics only make limited use of digital competences and digital technology and focus on clinical processes.

As a result, considering the differences in digital competences between faculties, it is of great importance to establish strategic approaches to the development of digital skills. In this context, the university administration can contribute to improving digital

competencies by designing faculty-specific training programs, increasing the effectiveness of teaching processes by strengthening the digital infrastructure and improving the digital skills of all academics through continuing education opportunities.

As with every study, this one has some limitations. Faculties' digital competence was assessed based on academics' self-assessment, which may limit the objectivity of the results. In addition, the study is limited to six faculties. Future studies conducted across the university and at different universities will increase the generalizability of the results.

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